

Chapter 16.

Feeding of Livestock and Poultry

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Feeding is a crucial aspect of livestock and poultry management. Proper nutrition ensures optimal growth, reproduction, and production of meat, milk, and eggs. Different species of animals have varying nutritional needs based on their physiological stages, breeds, and environmental conditions. This chapter explores the feeding requirements of livestock and poultry, types of feeds, feeding strategies, and best practices. Proper feeding of livestock and poultry is fundamental to their health and productivity. A well-balanced diet, combined with good management practices, ensures efficient animal production, economic sustainability, and food security. Farmers should adopt scientifically formulated feeds and feeding strategies to maximize output and profitability. India today, stands first in the area of milk production at the world level, with an annual growth rate of about 4% (Singh et. al., 2012). The productive improvements among dairying animals can be made through proper management, feeding and handling, etc (Singh et. al., 2013). Goat population of our country increased from 47.14 million in the year 1951 to 124.5 million during 2005 (Singh and Sharma, 2013a). Improvement can be made through proper management, feeding, handling and other environmental conditions which will influence expression of characters but a limit of which is set by heredity of individual (Singh et. al., 2013b). Goats are integral part of livestock production and play a vital role in the socio-economic structure of rural poor (Singh and Sharma, 2014). Various government and non-government organization have also recognized the importance of poultry farming as employment generating enterprise and are engaged in motivating more and more entrepreneurs to take up this enterprise (Singh et. al., 2014a). Goats play a vital socio-economic role in Asian agriculture, particularly for resource poor people living in harsh environment (Singh et. al., 2014b). The global goat population currently stands at 921 million, of which over 90% are found in

developing countries (Singh et. al., 2014c). This benefit is often not shown in national statistics because of informal trading and slaughtering (Singh et. al., 2014d). Goat milk contains less lactose than cow's milk, so is less likely to trigger lactose intolerance. The milk is naturally homogenized since it lacks the protein agglutinin (Singh et. al., 2014e). The goat was domesticated as early as 6-7 BC, as evidenced by archaeological remains collected in western Asia (Singh et. al., 2014f). Major population of India is primarily depends on agricultural based system for their daily life including goat keeping that constitute an important rural business of small marginal farmers and landless labours (Singh et. al., 2014g). Reproductive management of an animal is governed through a number of parameters, viz. age at first conception, age at first calving and first gestation length etc (Singh et. al., 2014h). Goat milk contains less lactose than cow's milk, so is less likely to trigger lactose intolerance (Singh and Sharma, 2015). It has since played a significant socioeconomic role in the evolvement of human civilization around the world (Singh and Sharma, 2015a). Farmers preferred Deda over Kona because it has more biomass (Singh and Sharma, 2015b). A very important aspect in this regard is the awareness of risk by resource-poor farmers and their emphasis on minimizing it (Singh and Sharma, 2016). The country is endowed with large and biologically diverse population of goats (Singh and Sharma, 2016a). The nutritional value of milk is closely related to its composition, which is affected by factors such as breed, diet, stage of lactation, season etc (Singh and Sharma, 2016b). Livestock production is backbone of Indian agriculture contributing 7% to national GDP and source of employment and livelihood for 70% population in rural areas (Singh et. al., 2017). Animals reared in intensive production systems consume a considerable amount of protein and other nitrogen-containing substances in their diets (Singh et. al., 2017a). Small ruminants have a large impact on the economy and food supply of people in subtropical and tropical countries (Singh and Sharma, 2017b). Goats play a vital socio-economic role in Asian agriculture, particularly for resource-poor people living in harsh environments (Singh et. al., 2025e). Non-cattle milk accounts for approximately 15% of the total milk consumption by humans worldwide (Singh et. al., 2025d). Asia contributes approximately 59% to world goat milk production (Singh et. al., 2018). Jamnapari (or Jamunapari) is a breed of goat originating from Indian subcontinent. Since 1953 they have been imported to Indonesia (popular as Etawa goat, and its mixture with a local goat called "PE", *peranakan Etawa* or Etawa mix) where they have been a great success (Singh et. al., 2025c). The name is derived from the rivers Yamuna, Jamuna (West Bengal) and

Jamuna (Bangladesh) of India and Bangladesh (Singh et al. 2025). There is a great variation in coat colour, but the typical coat is white with small tan patches on head and neck (Singh et. al., 2024e). The typical character of the breed is a highly convex nose line with a tuft of hair, yielding a parrot mouth appearance (Singh et. al., 2017c). The consequence of domestication was a change in the phenotypic characteristics of wild goats, which resulted in the development of a multiplicity of goat breeds or types (Singh et. al., 2024d). These breeds or types were distributed across the world as a result of the migration and translocation of humans, usually due to changing climatic conditions and natural resources (Singh and Sharma, 2017d). There is a large commercial chicken industry that provides us with eggs and meat (Singh, G. 2019). Milk-secreting tissues and various ducts throughout the udder can be damaged by bacterial toxins, and sometimes permanent damage to the udder occurs (Singh et. al., 2024c). Severe acute cases can be fatal, but even in cows that recover there may be consequences for the rest of the lactation and subsequent lactations (Singh and Singh, 2020). Livestock has become an integral part of all interventions aimed at reducing rural poverty and enhancing food and nutrition security (Singh et. al., 2025a). The dairy livestock owners who raise cattle and buffaloes are yet ignorant with scientific management practices (Singh and Somvanshi, 2020a). India is endowed with a significant share of the world's livestock population growing steadily and continuously (Singh et. al., 2025b). Buffalo is predominantly animal of poor countries with very high density of livestock and human population and with poor feed recourses (Singh et. al., 2024b). In tropical and subtropical regions dairy cattle usually depend exclusively on native or introduced pastures as their only source of nutrients, and in particular, during critical periods of the year, such as the winter or dry season, the animals cannot fulfill their nutrient requirements because forage is either scarce or of low quality (Singh, G., 2019a). The goat is thought to have been the earliest domesticated ruminant and of all the species of domesticated animals except dog, has the widest ecological range (Singh et. al., 2024a). Originating in Asia, goats have spread over all the continents and inhabit almost all -climatic zones from arctic - circle to the equator (Singh, G., 2024). Goat milk is rich in essential proteins, vitamins, and minerals (Singh et. al., 2023). Goat milk is widely consumed due to its high nutritional value and medicinal properties (Singh et. al., 2024f). Effective management practices, nutritional interventions, and adaptive breeding strategies can mitigate these challenges (Singh et. al., 2025e). The term Agriculture is derived from two Latin words ager or agri meaning soil and culture meaning cultivation (Singh, et., al.

2025g). The composition of mastitis milk approaches that of blood (Singh and Rodricks 2025). Feed alone constitute 60 per cent of the production cost of milk (Singh and Rodricks 2025a). The nutrient requirements are generally expressed separately for each function or an overall figure for the combined functions may also be expressed (Singh and Rodricks 2025b). The poultry industry has developed into a highly organized and scientific sector, contributing significantly to the global food supply (Singh, G. 2025q). The poultry industry in India is one of the fastest-growing sectors in agriculture, contributing significantly to food security, employment, and economic growth (Singh, G. 2025p). Livestock nutrition depends on a variety of feeds and fodders, which can be classified based on their composition, digestibility, and utility (Singh, G. (2025o). Livestock nutrition is a crucial aspect of animal husbandry, directly affecting growth, reproduction, milk production, and overall health (Singh, G. 2025n). Distributed in Salem, Erode, Karur, Namakkal, and fewer parts of Dharmapuri districts of Tamilnadu (Singh, G. 2025m). This is otherwise called as Delhi, Kundi and Kali (Singh, G. 2025l). A cattle farming is an integral part of Indian agriculture, providing milk, meat, draft power, and manure (Singh, G. 2025k). This breed is otherwise known as Desan, Gujarati, Kathiawari, Sorthi, and Surati (Singh, G. 2025j). Goats are the number one producer of milk in the world (Singh, G. 2025i). Many farmers in India depend on animal husbandry for their livelihood (Singh, G. 2025h). Mixed farming is an agricultural practice that combines crop cultivation with livestock rearing or other supplementary enterprises like fisheries, agroforestry, or poultry (Singh and Mishra 2025r). Fisheries play a vital role in India's economy, providing livelihood to millions, contributing to food security, and earning foreign exchange (Singh, G. 2025s). Fish production plays a significant role in global food security, employment, and economic development (Singh, G. 2025t). A person working with animals should have proper knowledge of the different parts of the animal body (Singh, G. 2025u). The weight of farm animals can be work out without weighing machine (Singh, G. 2025v). While taking work from farm animals or while milking, treatment, castration, applying identification mark (Singh, G. 2025w). If a herd's man has only few animals, recognizing each animal separately is possible for differentiating them according to their external appearance (Singh, G. 2025x). Ageing means to determine the approximate age of an animal (Singh, G. 2025y). Livestock feeds are generally classified according to the amount of specific nutrients they furnish in the ration (Singh, G. 2025z).

Feeding of Livestock

Livestock such as cattle, sheep, goats, and pigs require a balanced diet that includes carbohydrates, proteins, fats, vitamins, minerals, and water. The quality and quantity of feed directly impact their growth rate, health, and productivity.

1. Feeding of Cattle

Cattle can be categorized into dairy and beef cattle, each requiring a different feeding regimen.

- **Dairy Cattle:** Require high-energy and protein-rich diets to maintain milk production. They are commonly fed a combination of roughages (hay, silage, pasture) and concentrates (grains, oilseed meals, mineral supplements).
- **Beef Cattle:** Need diets that support muscle growth and fat deposition. They are typically fed forage-based diets supplemented with grains and protein sources to improve weight gain.

2. Feeding of Sheep and Goats

Sheep and goats are ruminants that rely primarily on forages such as grasses, shrubs, and hay. Their diet should include:

- High-quality roughages to support digestion and prevent bloating.
- Concentrates for enhanced growth and reproduction.
- Mineral supplements, especially calcium, phosphorus, and salt.

3. Feeding of Pigs

Pigs are monogastric animals with high energy and protein requirements. Their diet consists of:

- Energy-rich grains (corn, wheat, barley).
- Protein sources (soybean meal, fish meal, meat and bone meal).
- Vitamins and mineral supplements for proper growth and reproduction.
- Clean water for hydration and metabolic functions.

Feeding of Poultry

Poultry species such as chickens, turkeys, ducks, and quails require a nutritionally balanced diet to support egg production and meat yield. Their diet consists of:

- **Starter Feeds:** High-protein feeds for young chicks to promote growth.
- **Grower Feeds:** Moderate protein feeds to sustain development.
- **Finisher Feeds:** Energy-dense feeds for meat birds to enhance weight gain.
- **Layer Feeds:** Calcium-rich feeds for laying hens to support eggshell formation.
- **Supplementary Feeds:** Grits for digestion, vitamins for immune support, and probiotics for gut health.



Fig. Feeding of livestock, poultry and Swine

Types of Livestock and Poultry Feeds

1. **Roughages:** Grasses, hay, silage, and crop residues used mainly for ruminants.
2. **Concentrates:** Cereal grains, protein meals, and industrial by-products rich in nutrients.
3. **Supplements:** Minerals, vitamins, and additives that improve feed efficiency.
4. **Water:** Essential for digestion, metabolism, and body functions.

Feeding Strategies and Management Practices

- **Balanced Rationing:** Ensuring animals receive a diet that meets all nutritional needs.
- **Feed Formulation:** Mixing feeds to optimize nutrient intake and cost-effectiveness.

- **Proper Storage:** Preventing feed contamination by storing it in clean, dry conditions.
- **Regular Feeding Schedule:** Providing consistent feeding times to maintain animal health.
- **Disease Prevention:** Using quality feeds and clean water to reduce the risk of infections.

References

872. Singh, G., Dutt, G., Sharma, R.B., Fatima, A. and Singh, R.P. (2012). Study of first gestation length in Gir cows, *The Journal of Rural and Agricultural Research*, **12**(1): 64- 65.
873. Singh, G., Dutt, G., Sharma, R.B., Singh, S.K., Fatima, A. and Chauhan, S.V.S. (2013). An Analytical Study of Reproductive Performance in Gir Cows, *Indian Research Journal of Extension Education*, Special Issue, (2): 203- 206.
874. Singh, G. and Sharma, R.B. (2013a). Influence of breeds on goat milk composition under field and farm rearing conditions, *Indian Research. Journal of Genetics & Biotechnology*, **5**(4): 258- 261.
875. Singh, G., Dutt, G., Rajput, S. and Chauhan, R.S. (2013b). Study of age at first service period in Gir cows, *Indian Research. Journal of Genetics & Biotechnology*, **5**(4): 270- 273.
876. Singh, G. and Sharma, R.B. (2014). Effect of season on the milk quality of Jamunapari goats under field and farm rearing condition, *Indian Research. Journal of Genetics & Biotechnology*, **6**(1): 335- 339.
877. Singh, G., Thorat, G.N., Trivedi, M.S., Mishra, R. and Sharma, S.K. (2014a). A test to measure knowledge about poultry management practices, *The Journal of Rural and Agricultural Research*, **14**(2): 44- 47.
878. Singh, G., Sharma, R.B. and Mishra, R. (2014b). Seasonal variations in the milk minerals of Jakhrana goats under field and farm rearing conditions, *Journals of community mobilizations and sustainable development*, **9**(2): 120 – 123.
879. Singh, G., Sharma, R.B., Mishra, R. and Rajput, S. (2014c). Effect of multiple births on Jakhrana goat milk quality under field and farm rearing conditions, *Indian Research. Journal of Genetics & Biotechnology*, **6**(4): 629- 635.

880. Singh, G., Sharma, R.B., and Mishra, R. (2014d). Effect of season on the milk quality of Jakhrana goats under field and farm rearing condition, *Indian Research. Journal of Genetics & Biotechnology*, **6**(3): 571- 577.
881. Singh, G., Sharma, R.B., Mishra, R. and Rajput, S. (2014e). Effect of season on goat meat composition under field and farm rearing conditions, *Indian Research. Journal of Genetics & Biotechnology*, **6**(3): 511- 517.
882. Singh, G., Sharma, R.B. and Mishra, R. (2014f). Effect of multiple births on Jamunapari goat milk quality under field and farm rearing conditions, *Indian Research. Journal of Genetics & Biotechnology*, **6**(2): 453- 458.
883. Singh, G., Sharma, R.B., Kumar, A. and Chauhan, A. (2014g). Effect of Stages of Lactation on Goat Milk Composition under Field and Farm Rearing Condition, *Advances in Animal and Veterinary Sciences*, **2**(5): 287- 291.
884. Singh, G., Dutt, G. and Rajput, S. (2014h). Study of age at first calving in Gir cows, *Indian Research. Journal of Genetics & Biotechnology*, **6**(1): 362- 365.
885. Singh, G. and Sharma, R.B. (2015). Effect of multiple births on Jakhrana goat milk minerals under field and farm rearing Conditions, *Indian Research. Journal of Genetics & Biotechnology*, **7**(2): 227- 234.
886. Singh, G. and Sharma, R.B. (2015a). Influence of breed on goat meat composition under field and farm rearing Conditions, International conference on Emerging Trends in Biotechnology and Science with Especial Reference to Climatic Change, 15- 17 Feb., 2015 held at KVK Tonk Banasthali Vidyapith
887. Singh, G. and Sharma, S.K. (2015b). On Farm Trial (OFT) of pearl millet green fodder at Tonk district, *The Journal of Rural and Agricultural Research*, **15**(2): 28- 29.
888. Singh, G. and Sharma, R.B. (2016). Impact of stages of lactation on the minerals of Jakhrana goat milk under field and farm rearing condition, *Research Journal of Animal Husbandry and Dairy Science*, **7**(1): 28- 34.
889. Singh, G. and Sharma, R.B. (2016a). Effect of Goat Breeds on the Milk Mineral Composition under Field and Farm Rearing Conditions, *The Bioscan*, **11**(2): 691- 694
890. Singh, G. and Sharma, R.B. (2016b). Effect of rearing systems on mineral contents of milk during lactation in Jamunapari goats, *Indian Journal of Small Ruminants*, **22**(2): 270- 271.

891. Singh, G., Sharma, R.B. and Singh, M. (2017). Green Fodder Production Potential of Oat cv. Kent under Semi-arid Climatic Conditions of Tonk-Rajasthan in Frontline Demonstration, *International Journal of Current microbiology and Applied Sciences*, **6**(3): 2228- 2232.
892. Singh, G., Sharma, R.B. Singh, M. and Sharma, S.K. (2017a). Utilisation of agricultural wastes in participatory poultry farming with women under climatic conditions of Tonk district of Rajasthan, *Agricultural Science Digest*, **37**(1): 60- 63.
893. Singh, G. and Sharma, R.B. (2017b). Effect of Field and Farm Rearing Conditions on the Sensory Quality of Goat Meat, *Journal of Community Mobilization and Sustainable Development*, **11**(2):188- 192.
894. Singh, G., Sharma, R.B., Singh, M. and Choudhary, R. (2017c). Effect of season on jamunapari goat meat composition under field and farm rearing condition, *Indian Journal of Pure & Applied Biosciences*, **5**(2): 563-568.
895. Singh, G. and Sharma, R.B. (2017d). Seasonal impact on the minerals of jamunapari goat milk minerals under field and farm rearing condition, *International Journal of Current microbiology and Applied Sciences*, **6**(9): 1298- 1303.
896. Singh, G., Sharma, R.B., Chahal, B.P., Singh, M. and Sharma, S.K. (2018). Effect of multiple births on Jamunapari goat milk minerals under field and farm rearing conditions, *Indian Journal of Animal Research*, **52** (4): 628- 631.
897. Singh, G. (2019). Analytical study of Front Line Demonstration (FLD) of Kadaknath Poultry Farming under climatic conditions of Tonk District of Rajasthan, *The Journal of Rural and Agricultural Research*, **19**(2): 49- 52.
898. Singh, G. (2019a). Effect of area specific mineral mixture on productive performance of murrah buffaloes under climatic conditions of tonk district, *Indian Research. Journal of Genetics & Biotechnology*, **11**(4): 277- 281.
899. Singh, G. and Singh, R.P. (2020). An Analytical Study on Mastitis in Cows under Climatic Conditions of Tonk district, *The Journal of Rural and Agricultural Research*, **20**(1): 18- 21.
900. Singh, G. and Somvanshi, S.P.S. (2020a). Study on Animal Rearing Practices by Dairy Owners of District Jaipur Rajasthan, *The Journal of Rural and Agricultural Research*, **20**(1): 61- 64.

901. Singh, G., Kumar, A., Chauhan, R.S., Mishra, A.K., Singh, G., Sharma, L.K., Kumar, S., Dhayal, L.S. and Goura, R.S. (2023). Study of medicinal properties of goat milk on physiological disorders in human beings at Agra district, Uttar Pradesh, *Journal for ReAttach Therapy and Developmental Diversities*, **6**(10s) (2): 1852- 1855
902. Singh, G. (2024). Nutrition and feeding management of goats for chevon production, *International Journal of Science, Environment and Technology*, **13**(5): 334-349.
903. Singh, G., Singh, S., Sharma, K., Sharma L.K. and Kumar, A. (2024a). Effect of goat rearing on environment and rural prosperity in India, *International Journal of Science, Environment and Technology*, **13**(6): 421- 433.
904. Singh, G., Bhati, D.S., Sharma, K. and Kumar, N. (2024b). Effect of goat breeds on the milk composition under climatic conditions of Baijupara tahsil of Dausa district Rajasthan, *Journal of Progressive Agriculture*, **15**(2): 49- 60.
905. Singh, G., Sharma, K., Sharma L.K., Kumar, A. and Parihar, K. (2024c). OFT (On Farm Testing) on the Area Specific Mineral Mixture on the Milk Production of Murrah Buffaloes under Climatic Conditions of Tonk District, *The Journal of Rural and Agricultural Research*, **24**(2): 11- 17.
906. Singh, G., Sharma, K., Sharma L.K., Kumar, A. and Parihar, K. (2024d). FLD (Front Line Demonstration) on the Area Specific Mineral Mixture on the Milk Production of Murrah Buffaloes under Climatic Conditions of Tonk District, *The Journal of Rural and Agricultural Research*, **24**(2): 51- 57.
907. Singh, G., Sharma, K., Sharma L.K., Kumar, A. and Kumar, N. (2024e). An analytical study on bloat in buffaloes under climatic conditions of Tonk district of Rajasthan, *The Journal of Rural and Agricultural Research*, **24**(2): 76- 81.
908. Singh, G., Kumar, A., Arha, A., Mishra, A.K., Kumar, A., Singh, G., Kumawat, B. and Kumawat, P., Chauhan, G.S. (2024f). Study of medicinal properties of goat milk on physiological disorders in human beings at Dausa District, Rajasthan, *Cuestiones de fisioterapia*, **53**(3):644-651
909. Singh, G., Sharma, K., Tandon, C., Pandya, P., Verma, A. and Kumar, N. (2025). Effect of goat breeds on the milk composition under climatic conditions of Bhandarej

tahsil of Dausa district Rajasthan, *Asian Journal of Advances in Agricultural Research*, **25**(1): 10- 18.

910. Singh, G., Sharma, K., Tandon, C., Pandya, P., Verma, A. and Kumar, N. (2025a). Effect of goat breeds on the milk composition under climatic conditions of Lalsot tahsil of Dausa district Rajasthan, *International Journal of Agriculture Extension and Social Development*, **8**(1): 144- 149.
911. Singh, G., Chauhan, R.S., Kumar, A., Sharma, K., Kumar, N., Swarankar, P.K and Goura, R.S. (2025b). Influence of seasons on the goat milk composition reared under conditions of Amber tehsil of Jaipur district, *International Journal of Geography, Geology and Environment*, **7**(1): 21- 28.
912. Singh, G., Chauhan, R.S., Kumar, A., Sharma, L.K., Rodricks, C.C., Kumar, D., Kumar, N., Singh A.P., Tandon C., Prince, K. Pandya, P. and Kumawat, P. (2025c). Effect of goat breeds on the milk composition under climatic conditions of dausa tehsil of dausa district rajasthan, *International Journal of Science, Environment and Technology*, **14**(1), 1- 14.
913. Singh, G., Mahesh, M.S., Parkash, J., Somvanshi, S.P.S., Kumar, A., Singh, G., Kumar, A., Sharma, L.K., Shalini, Purohit, H., Goura, R.S. (2025d). Influence of seasons on the composition of goat milk reared under conditions of mauzmabad tehsil, Jaipur district, *International Journal of Agriculture and Nutrition*, **7**(2): 09-14.
914. Singh, G., Parkash, J., Somvanshi, S.P.S., Kumar, A., Singh, G., Kumar, A., and Goura, R.S. (2025e). An analytical study on Foot and Mouth Disease under climatic conditions of Tonk district, *International Journal of Veterinary Sciences and Animal Husbandry* 2025; **10**(2): 225-229.
915. Singh, G., Sharma, K., Singh, G., Kumawat, B., Kumawat, P., Kumar, A., Kumar, S., Mishra, A.K., Goura, R.S. and Dhayal, L.S. (2025f). Effect of Environmental Factors on the Milk Composition of Goats under Chaksu Tehsil of Jaipur District, *The Academic*, **3**(2):520 – 528.
916. Singh, G., Sharma, V. K. and Prince, K. (2025g) Introduction of Indian Agricultural Heritage, A book, *Agricultural Heritage*, 'ISBN. No. 978-81-8268-238-2, pg., 01 – 17.

917. Singh, G. and Rodricks, C.C. (2025). Clean Milk Production and processing, A book --*'Dairy Cattle and Buffaloes Production and Management'*, ISBN.No. 978-81-8268-225-2, pg., 256 – 273.
918. Singh, G. and Rodricks, C.C. (2025a). Feed and Fodder Management, A book entitled *'Dairy Cattle and Buffaloes Production and Management'* ISBN. No. 978-81-8268-225-2, pg., 65 – 118.
919. Singh, G. and Rodricks, C.C. (2025b). Concept of Indian Feeding standard, A book entitled *'Dairy Cattle and Buffaloes Production and Management'*, ISBN. No.978-81-8268-225-2, pg., 65 – 118.
920. Singh, G. (2025h). Animal husbandry methods in India, A book entitled, *'Livestock and Poultry Management'*, ISBN. No. 978-93-342-6054-0, pg. 24 – 31.
921. Singh, G. (2025i). Common terms pertaining to different species of livestock, A book entitled *'Livestock and Poultry Management'*, ISBN. No. 978-93-342-6054-0, pg. 32 – 41.
922. Singh, G. (2025j). Utility classification of breeds of cattle and Buffaloes, A book entitled *'Livestock and Poultry Management'*, ISBN. No. 978-93-342-6054-0, pg. 42 – 78.
923. Singh, G. (2025k). Familiarization with different breeds of cattle (indigenous and exotic) with special emphasis on breeds of Rajasthan, A book entitled *'Livestock and Poultry Management'*, ISBN. No. 978-93-342-6054-0, pg. 79 – 86.
924. Singh, G. (2025l). Familiarization with different breeds of buffaloes with special emphasis on breeds of Rajasthan, A book entitled *'Livestock and Poultry Management'*, ISBN. No. 978-93-342-6054-0, pg. 87 – 94.
925. Singh, G. (2025m). Classification of breeds of sheep and goat, A book entitled *'Livestock and Poultry Management'*, ISBN. No. 978-93-342-6054-0, pg. 95 – 126.
926. Singh, G. (2025n). Introduction to common feeds and fodders, A book entitled *'Livestock and Poultry Management'*, ISBN. No. 978-93-342-6054-0, pg. 127 – 133.
927. Singh, G. (2025o). Classification and utility of common feeds and fodders, A book entitled *'Livestock and Poultry Management'*, ISBN. No. 978-93-342-6054-0, pg. 134 – 140.

928. Singh, G. (2025p). Introduction to poultry industry in India (past, present and future status), A book entitled '*Livestock and Poultry Management*', ISBN. No. 978-93-342-6054-0, pg. 141 – 140.
929. Singh, G. (2025q). Common terms pertaining to poultry production and management, A book entitled '*Livestock and Poultry Management*', ISBN. No. 978-93-342-6054-0, pg. 147 – 153.
930. Singh, G. and Mishra, A.K. (2025r). Concept of mixed farming and its relevance to socio-economic conditions of farmers in India, A book entitled '*Livestock and Poultry Management*', ISBN. No. 978-93-342-6054-0, pg. 154 – 161.
931. Singh, G. (2025s). Importance of fisheries in India, A book entitled '*Livestock and Poultry Management*', ISBN. No. 978-93-342-6054-0, pg. 170 – 177.
932. Singh, G. (2025t). Importance of fisheries in India, A book entitled '*Livestock and Poultry Management*', ISBN. No. 978-93-342-6054-0, pg. 170 – 177.
933. Singh, G. (2025u). Study of body parts and points of cattle, sheep, goat and their significance, A Practical Manual entitled '*Principles of Livestock Production and Management*', ISBN No. 978-93-342-6199-8, pg. 01- 19.
934. Singh, G. (2025v). Measuring and Weighing of Farm Animals, A Practical Manual entitled '*Principles of Livestock Production and Management*', ISBN No. 978-93-342-6199-8, pg. 20- 32.
935. Singh, G. (2025w). Use of Common Restraints Used in Different Animals, A Practical Manual entitled '*Principles of Livestock Production and Management*', ISBN No. 978-93-342-6199-8, pg. 33- 46.
936. Singh, G. (2025x). Systems of Identification of Livestock, A Practical Manual entitled '*Principles of Livestock Production and Management*', ISBN No. 978-93-342-6199-8, pg. 47- 62.
937. Singh, G. (2025y). Methods of Determination of Age in Farm Animals, A Practical Manual entitled '*Principles of Livestock Production and Management*', ISBN No. 978-93-342-6199-8, pg. 63- 78.
938. Singh, G. (2025z). Identification of Common Feeds and Fodders, A Practical Manual entitled '*Principles of Livestock Production and Management*', ISBN No. 978-93-342-6199-8, pg. 79- 102.

